STATEMENT OF WORK

AI & Programmatic Decision Platform

Prepared for:

Rosotravel.com

Prepared by:

Appinventiv Technology

Version 2.1 Revised per Client Feedback

Date: May 2026

Confidential For Client Review Only

1. Document Control

Client	Rosotravel.com
Service Provider	Appinventiv Technology
SOW Scope	AI & Programmatic Decision Platform original scope
Version	2.1 Revised per client feedback dated May 2026
Project Scopes	This SOW covers the AI deliverables within 2. Scope: Programmatic Decision Platform, SEO & Website Architecture. Items outside this AI scope are identified in Section 5. Where items reference frontend/UX delivery, this falls under the separate UX & Design scope. Payment processor (Chapter VI) and partner programs (Chapter VII) are chapters within Scope 2 outside this AI SOW.
Delivery Model	Phased delivery within original scope see Section 3 for phase definitions
Target Completion	December 2026

Version 2.1 incorporates all feedback from the client commentary document dated May 2026. Changes from v1.0 include: corrected destination ingestion model, updated audience taxonomy (7 audiences), added intent/constraint/eligibility classification, added Model C product selection logic, decision proof content, two-stage near-page model, review loop foundations, Human Face / Snapshot foundations, city/country brand expertise blocks, CRM-ready data layer foundations, DEE boundary table, media rights governance, atomic publishing rules, external API failover, and corrected out-of-scope ownership mapping.

2. Executive Summary

Rosotravel.com is repositioning from a traditional tour operator into a decision-first, programmatic experience platform The World's Most Recommended Experience Expert. This SOW defines the AI and programmatic platform scope (original scope) that powers this transformation.

The platform serves three audiences simultaneously from a single source of truth: Google crawlers (programmatic SEO at scale), LLM systems such as ChatGPT and Perplexity (structured feeds and MCP), and human travelers (curated, decision-confident experience pages). The core system is not AI as a brand promise it is a governed, deterministic architecture where AI is the production engine, humans retain control through locks and QA, and the Decision Platform logic (Model C) ensures curated quality at scale.

Target outcome by December 2026: minimum 300,000 organic sessions per month across Google and LLM referrals and AI agent bookings, conversion rate above 1.0%, SEO ROI above 300%, 100% of new or changed tours published within 24–48 hours.

3. Delivery Phases

Important clarification: The original scope specification does not define a single unified project-level Phase 1/2/3 delivery sequence. Phases in the spec are feature-level roadmap markers within specific modules (MCP, DEE, media). The phasing below is Appinventiv's proposed delivery sequence to be confirmed and baselined with the client during the discovery workshop. All phases are within original scope unless explicitly stated otherwise.

Phase	What is delivered	Key modules
MVP	Core AI pipeline: destination ingestion, RAW store, Diff Engine, factual enrichment, media classification (Class A/B/C), Cloudinary, keyword retrieval, Qdrant (4 collections), embedding pipeline, LangChain 4-chain drafting, schema builder, FAQ pipeline, validation layer, human QA, atomic publishing, translation + hreflang, internal linking outputs, CMS panels (Marketing + Product Ops + SEO governance + Optimization Inbox + RBAC), LLM visibility feeds (llms.txt, ai-sitemap.xml, ai-summary.json, inventory feed), Model C classification engine (RAW→Pool→Set, archetype winner logic, rails, decision proof), external API failover + cost governance, MCP server (3 tools: search_tours + get_tour + get_availability_link), Snapshot / Human Face / Trip DNA foundations, review loop foundations	Systems 1, 2, 3, 4, 6, 7 (MVP tools), 10. Spec refs: Section 1 overview lines 1013–1018, MCP Phasing Phase 1, DEE Roadmap Phase 1 (MVP), Snapshot Section 39 MVP, Review loop Section IX, Media pipeline Phase 1
Phase 1.5	Nano Banana Pro image generation for Class A assets, ALT text automation for processed assets, schema exposure for processed images, admin-uploaded Snapshot media, stronger city/attraction-level snapshot aggregation, improved proof labels by source confidence, better guide anchor linking	Systems 1 (extension), 5 (extension). Spec refs: Media pipeline Phase 1.5, Snapshot Section 40 Phase 1.5
Phase 2	DEE variant generation (7 audiences, Chain 4) + CMS fields, Stage 1 immediate near-pages (batch-time), Stage 2 post-publish near-pages (5-week gate), continuous optimization loop (daily GSC/GA4 + weekly LangChain), public API documentation (OpenAPI + MCP manifest), get_bundles MCP tool, Agentic Commerce API (Quote +	Systems 2 (extension), 5, 7 (Phase 2 tool), 8, 9. Spec refs: DEE Roadmap Phase 2, MCP Phasing Phase 2, Media pipeline Phase 2 (Veo video for cleared assets)

	CheckoutSession + Order), headless checkout UX, agent/LLM discount, CRM-ready data layer (all events + identity + consent + webhooks), rating aggregation (product/city/country), review request eligibility trigger, city/country brand expertise blocks, persistent DEE profiles (opt-in), richer review theme summaries per audience, expanded near-page theme library, bundle selection logic	
Phase 3 (conditional)	Agent-assisted planning handoff, predictive intent improvements (governed), seasonal intent shifts, deeper personalization across journeys, hero video generation with Google Veo (conditional on Viator licensing confirmation), expansion of processed/indexable supplier media (conditional on partner license review)	Systems 2 (extension), 5 (extension), 7 (Phase 3). Spec refs: DEE Roadmap Phase 3, MCP Phasing Phase 3, Media pipeline Phase 3 conditional

The testing environment (rosotravel.at subdomain) is used for all phase validation. One complete city batch is published and fully tested on rosotravel.at before any migration to rosotravel.com. All three phases are in original scope. Phase boundaries and timelines are to be agreed during discovery.

4. AI Scope Systems

System 1: Data Ingestion & Versioning Engine

Purpose

Ingest all entity data from internal and external sources into a governed RAW store, version every record, detect changes with hash precision, and prevent unnecessary AI reprocessing.

1.1 Destination Hierarchy Ingestion

> *Correction from v1.0: Destination hierarchy is NOT supplied as a manual marketing file. It is ingested, refreshed, and maintained from the Viator API feed where available. Rosotravel then internalises, maps, approves, merges, categorises, and assigns stable internal entity_ids and codes via the CMS. This is a platform data architecture requirement, not a manual dependency.*

The destination hierarchy (Country > Region > City) (NOTE: destination name mapping and normalisation from Viator's naming conventions will be detailed during the discovery phase.)is ingested from the Viator API destination feed and normalised into a canonical English form. Rosotravel administrators review, approve, merge duplicates (An automated layer of duplication detection will be needed), and assign stable internal entity_ids in the CMS. A destination becomes active once at least one product is linked to it. Active destinations are published. Inactive destinations are never indexed. The hierarchy refreshes on a configured cadence via API not via manual file upload (API updates **cannot overwrite or delete** an existing approved destination without explicit CMS acceptance - the destination tree is protected).

1.2 Product Ingestion

- External channel: Viator / OTA API feeds ingested automatically on a configurable cadence
- Internal channel: products added manually via CMS by the Rosotravel operations team
- Internal products: receive a one-time AI rewrite on first ingestion, then become manual-only. Any field can be manually edited and locked permanently. AI never overwrites locked internal product fields.
- External products: AI rewriting is allowed on each MAJOR diff event unless a specific field has been manually edited, in which case that field is locked permanently and skipped by AI on all future runs. Unlocking requires explicit CMS action with reason entry and audit log.

1.3 Diff Engine

Every new RAW snapshot is compared against the previous version using hash-based comparison across five domains:

Hash Domain	What It Covers	Severity & Action
content_hash	Title seed, description seed, highlights	MAJOR full AI rewrite queued

factual_hash	Hours, address, geo, phone, ratings	MAJOR schema refresh triggered
media_hash	Images, video references	MEDIUM media pipeline re-runs
offer_hash	Options, inclusions, cancellation	MEDIUM offer section updated
realtime_offer_hash	Price, availability slots	MINOR updates instantly, no AI

Hard rule: Only MAJOR severity triggers AI drafting. MINOR and MEDIUM bypass AI entirely. This protects cost and prevents content churn.

1.4 Factual Enrichment

- Google Places API: opening hours, formatted address, lat/lng coordinates, international phone, official website, rating, review count stored in relational DB, not Qdrant
- Wikidata API: inception year, architectural style, entity type, sameAs identifiers, isPartOf/hasPart relations, fallback coordinates
- Factual data is stored with its own factual_hash. Changes trigger schema refresh but not AI content regeneration unless the dependent block is explicitly configured to regenerate.
- AI never fabricates coordinates, opening hours, phone numbers, or links these always come from the factual layer.

1.5 Media Pipeline & Rights Governance

> *Media rights governance is a legal and SEO-safety requirement. The spec defines three media source classes. All pipeline rules derive from source class assignment.*

Class	Source	What is allowed
Class A Rosotravel-owned / cleared	Rosotravel originals, guide/expert images, admin-uploaded, customer review media where terms grant reuse rights	May be processed by Nano Banana Pro, used as indexable, assigned AI-generated ALT text, exposed in ImageObject / VideoObject schema
Class B Viator-sourced / restricted	Supplier photos from Viator API, review-linked traveler photos, other license-restricted assets	Performance optimisation only (WebP, CDN, lazy-load). MUST remain non-indexable. MUST NOT receive SEO alt text. MUST NOT be processed by Nano Banana unless explicit written Viator permission is obtained.
Class C AI-derived	Assets processed or generated by Nano Banana Pro or Google Veo from a source asset	Inherits rights risk of source asset. Only indexable and schema-eligible if source is Class A.

Every media asset in the system carries a source_class field (A / B / C), rights_status (owned / cleared / restricted / unknown), and indexable flag (true / false). These fields control alt text generation, schema exposure, Nano Banana processing eligibility, and sitemap inclusion. Restricted or unknown assets default to non-indexable.

Recommended rollout: Phase 1 keep Viator-sourced images non-indexable, use Class A assets as indexable, enable ALT text only for Class A. Phase 2 enable Nano Banana for Class A product assets, ALT text automation, schema exposure. Phase 3 (conditional) expand to Viator-derived media only after explicit written licensing confirmation.

System 2: Programmatic SEO Pipeline

Purpose

Auto-generate, validate, publish, and continuously optimise hundreds of thousands of SEO-compliant pages across all entity types, with zero keyword cannibalization, full schema compliance, and deterministic content quality at scale.

2.1 Keyword Retrieval & Embedding

- SEMrush API: volume, keyword difficulty, CPC, intent classification (navigational, informational, commercial, transactional, mixed), People Also Ask questions
- Google Search Console API: real user queries, impressions, CTR, position, emerging long-tail queries
- All keywords normalised before embedding: lowercased, punctuation-stripped, stopword-normalised, UTF-8 normalised
- Embedded using OpenAI text-embedding-3-large (3072-dimension float32 vectors)
- SEMrush unit caps, endpoint allowlists, batching strategy, and refresh frequency controls are implemented to prevent runaway API costs. Unnecessary re-retrieval is blocked by keyword_version tracking.

2.2 Qdrant Vector Database (4 Collections)

Collection	What Is Stored	Purpose
keywords_vectors	Keywords + 3072-dim vectors + ownership locks + version	Anti-cannibalization, similarity gate before keyword assignment
pages_vectors	Published pages + vectors + CTR/impressions + entity_type	Duplicate detection, related tours, internal linking suggestions
faq_vectors	FAQ questions + vectors + owner page	Global FAQ deduplication one owner per question
poi_vectors	Attractions + vectors + lat/lng + entity_type	Nearby attractions, global map, LLM dataset

Qdrant stores only embeddings and semantic metadata. All factual data (opening hours, addresses, coordinates as ground truth) lives in the relational database. Qdrant answers similarity questions; the relational DB answers factual questions.

2.3 Similarity Thresholds (Hard Rules)

Cosine Similarity	Rule
≥ 0.92	DUPLICATE blocked. No new URL, no new page.

0.78 – 0.92	VARIANT attaches as secondary keyword to existing owner. No new URL.
0.62 – 0.78	BORDERLINE human decision: absorb as FAQ section into existing page, or create near-page if gates pass.
< 0.62	NEW TOPIC eligible for a new page, subject to page-type routing and intent allowlists.
Pages: ≥ 0.90	Duplicate page blocked from publishing.
Pages: 0.70 – 0.90	Must be merged into existing owner before publishing.
FAQ: ≥ 0.85	Duplicate FAQ question blocked. Original owner retains the question.

2.4 LangChain Keyword Clustering & Allocation

- Product pages: transactional and bottom-funnel keywords
- Attraction / POI pages: mid-funnel commercial keywords
- City and Things-to-Do pages: broad transactional and commercial keywords
- Immediate near-pages: long-tail commercial clusters validated during batch processing (Stage 1 see Section 2.8)
- Post-publish near-pages: real GSC emerging queries at 5-week gate (Stage 2 see Section 2.8)
- Blog: informational-only keywords topic pool generated by system, article body written manually by expert writers

Once a keyword is allocated and locked, it cannot be reassigned. The keyword_map table records every assignment permanently with timestamp, batch_id, and assigning agent. Topic Graph is written after clustering: every entity node carries topic_id, primary_keyword, secondary_keywords[], representative_embedding, assigned_entity_url, parent_topic, children_topics[], and relationship types (isPartOf, hasPart, isSimilarTo, overlapsWith, nearby, related). No page may be generated, indexed, or exposed to LLMs unless it exists as a valid Topic Graph node.

2.5 LangChain 4-Chain AI Drafting Pipeline

- Chain 1 Retrieval: pulls keyword_map, factual data from relational DB, Qdrant page/FAQ/POI context, entity relationship graph. Assembles raw_data_enriched.json as the complete input.
- Chain 2 Drafting: sends enriched context + locked keyword map + prompt template to OpenAI GPT. Outputs: H1, H2s, H3s, meta title (60–75 chars), meta description (140–160 chars), full body, highlights, itinerary overview, pro tips, alt text for Class A images only, LLM snippets (short_summary for ai-summary.json), and full JSON-LD schema with URL placeholders.
- Chain 3 Refinement: GPT consistency validation (factual accuracy, naturalness, SEO compliance, no unsupported claims), Copyscape plagiarism scan (English product pages), Screaming Frog N-gram scanning (template repetition / AI footprint). Any failure routes back to Chain 2 with adjusted instructions.
- Chain 4 Variant Generation: generates 7 audience-specific snippet variants per page for DEE (see Section 2.6). Stored in CMS. Never generated at runtime.

2.6 Audience Taxonomy & DEE Variant Generation

Correction from v1.0: The audience taxonomy has been updated to 7 audiences. couple_traveler is added. family_planner is corrected to family_traveler. first_time_visitor remains the default.

Every page that has snippet variants stores exactly 7 variants one per audience:

- snippet_first_time_visitor (default rendered when no signal is detected)
- snippet_family_traveler
- snippet_couple_traveler
- snippet_comfort_easy_pace_traveler
- snippet_solo_social_traveler
- snippet_interest_deep_dive_traveler
- snippet_active_adventure_traveler

All 7 variants are pre-written during AI drafting and stored as CMS fields. DEE selects among them at runtime based on signal groups (G1 explicit > G2 landing context > G3 query/routing > G4 on-site behaviour > G5 locale). DEE never generates content at runtime and never creates new indexable URL variants.

CMS fields per variant: text content, lock flag, regenerate trigger, revert-to-AI option, last_modified_by. Fallback rule: if no signal detected or signal confidence is below threshold > render first_time_visitor variant.

2.7 Intent, Constraint & Eligibility State Classification

New in v2.1: This classification layer is part of the Model C taxonomy and governs how products are tagged, ranked, and rendered across all page types.

Intent **(0..2 active per session):**

- best_value, premium_comfort, hidden_gems, less_crowded, bucket_list_icons, deep_dive_learning, activity_intensity, family_compatibility, private_upgrade

Constraints **(0..n active; some are hard filters):**

- private_only (hard filter no non-private products may appear when active)
- small_group_preferred, low_walking_required, stairs_sensitive, stroller_required, wheelchair_access_needed
- short_time, half_day_only, hotel_pickup_preferred, free_cancellation_preferred, strict_budget_cap, weather_sensitive

Eligibility State **(per product, per context):**

- strong_fit product matches audience + intent + constraints well
- weak_fit product partially matches but has notable gaps
- unknown insufficient data to classify confidently
- not_recommended product does not meet context requirements

Hard rules: private_only is a hard filter non-private products are excluded entirely when this constraint is active. not_recommended products cannot win any main-stack slot. unknown never gives a ranking advantage. hidden_gems must not be merged with adventure or activity_intensity intents they are distinct positioning clusters.

Constraints map from explicit user signals (quiz, filter chips, accessibility selectors) and keyword patterns (e.g. 'wheelchair accessible Rome tour' > wheelchair_access_needed constraint active). Intent maps are additive (0..2 intents can be simultaneously active).

2.8 FAQ Pipeline

1. Retrieve FAQ-intent questions from SEMrush Questions, People Also Ask, and real GSC queries
2. Embed all candidates using OpenAI text-embedding-3-large
3. Query Qdrant faq_vectors similarity ≥ 0.85 to an existing owned question > blocked
4. Assign ownership (one owner globally per question, permanent)
5. OpenAI generates answers using locked question, page context, factual data (opening hours, practical info)
6. Human QA sampling reviews FAQ answers as part of the 3–5% batch sample

FAQ counts: Product pages 10, City pages 9, Attraction pages 7, Country/Region/Things-to-Do/Near-Page/Blog 5–7. FAQPage and HowTo schema emitted ONLY when the matching UI block is visible on the page. Schema must match visible content exactly UI/schema mismatch blocks publishing.

2.9 Schema & Structured Data

Schema is generated by a Schema Builder Engine AI cannot fabricate schema values. All factual values come from the factual layer. Required metadata policy per entity:

- @id: stable, permanent, never reused or regenerated on minor edits. Format: canonical URL with entity type prefix.
- datePublished: set at first publish. Never updated.
- dateModified: updated on every content or schema change. Shown in UI only if different from datePublished.
- lastReviewed: updated when human editor reviews or locks a page.
- author / expert attribution: included where an expert or guide is associated with a page.
- Validation gate: schema errors block publishing. Missing @id, datePublished, dateModified, or H1/title on any language > publishing blocked.

schema_map is maintained per page type every page type has an explicit list of required and optional schema types. faq_map maintains global FAQ ownership across the domain. AggregateRating schema is emitted ONLY when the rating value is visible in the UI and the schema value matches the displayed value exactly.

2.10 Validation Layer

- GPT consistency validation: factual accuracy, logical flow, naturalness, SEO compliance, no unsupported or misleading claims
- Over-optimisation / repetition check: N-gram scanning via Screaming Frog AI footprint detection, keyword stuffing detection, template-sameness across batch
- Copyscape plagiarism scan: English product pages

- Metadata completeness gate: title, meta description, H1, canonical, hreflang (all languages) missing any > batch held
- UI-to-schema parity gate: AggregateRating schema must match visible rating. FAQPage schema only when FAQ block is rendered. Any mismatch > page blocked.
- JSON schema validation: all JSON-LD validated against schema.org specification before publishing
- Factual integrity gate: coordinates, hours, phone validated against factual layer AI-fabricated factual fields are rejected

2.11 Human QA Sampling

3–5% of every batch is selected for manual review by an SEO specialist. Criteria: tone, factual accuracy, keyword usage, decision-platform content quality (not just SEO descriptions), style consistency, no misleading claims. If sample passes > full batch proceeds. If sample fails > prompts or keyword rules are adjusted and batch regenerates. Gate applies to every batch without exception.

2.12 Publishing Layer & Atomic Guarantees

> *New in v2.1: Publishing is atomic and queue-safe. Partial failures cannot create broken indexable pages.*

- Published Record assembly uses deterministic winner rule: human-locked content wins over AI-generated content, which wins over RAW seed content. Supplier raw payloads never appear in the Published Record.
- Failed AI jobs go to retry queue (3 attempts, exponential backoff), then dead-letter queue with CMS alert. Failed jobs never result in a partially published page.
- Long-running jobs run asynchronously. Page publishing is held until all constituent jobs complete successfully.
- Daily publication caps enforced to prevent crawl budget spikes and Google downgrade risk. Caps configurable per entity type.
- Sitemap.xml updated. ai-sitemap.xml updated. ai-summary.json regenerated. All published pages re-embedded into Qdrant pages_vectors after successful publish.
- Translation (OpenAI, semantic preservation) runs after English publish. hreflang tags generated and validated for correctness (every hreflang URL must resolve to a 200, canonical must point to correct language version, x-default must be set). Lock inheritance across languages: if a field is locked in English, it is locked in all translated versions until explicitly overridden per language.
- NER (Named Entity Recognition) preservation during translation: place names, product names, brand names, and proper nouns are preserved in their canonical form and not translated where the original form is correct. NER dictionary is configurable per language in CMS.
- Localized URLs where applicable: the URL architecture supports language-prefixed slugs (e.g. /en/italy/rome/, /it/italia/roma/, /de/italien/rom/) where the target language has a distinct canonical slug form. URL slug translation rules are defined per language in the implementation spec and enforced at publishing. All language variants carry hreflang and point to each other as alternates.
- CMS localization review and override fields: every translated page exposes per-language CMS fields for editor review and independent override. Fields:

lang_content_status (auto_translated / editor_reviewed / editor_locked), lang_lock_override (boolean, per language), lang_last_reviewed_by, lang_last_reviewed_at. A field locked in English can be independently unlocked and overridden in a specific language without affecting other languages or the English source.

- Localized content embeddings: after translation, each language version of a published page is embedded into Qdrant pages_vectors using the same OpenAI text-embedding-3-large model applied to the translated text. This ensures semantic clustering, duplicate detection, and related-content suggestions function correctly in non-English markets not just in English.

2.13 Internal Linking & Relation Outputs

Topic Graph relation outputs are exposed to frontend page templates through the Published Record. Every published entity includes a relations block with:

- isPartOf: parent canonical URLs (Country > Region > City hierarchy)
- hasPart: children canonical URLs (City > Attraction > Product)
- nearby: geo-derived canonical URLs (Qdrant poi_vectors, configurable radius)
- related: semantically similar pages (Qdrant pages_vectors, configurable threshold)
- internal_links_suggested: LangChain-generated internal link recommendations for body content

These relation fields drive the frontend 'Nearby Attractions', 'Related Tours', and 'You May Also Like' blocks. The routing table (Model C whole site) defines which relation types are rendered on which page templates. Appinventiv will convert the routing table from the original scope specification into implementation-ready logic during discovery.

2.14 Two-Stage Near-Page Generation

> *Correction from v1.0: Near-pages exist in two distinct stages, both of which are in scope.*

Stage 1 Immediate Batch-Time Near-Pages:

- Created during initial batch processing when long-tail keyword clusters are already validated by SEMrush volume and intent data
- No 5-week wait these are created at the same time as the main batch
- Must pass: similarity gate (< 0.62 from all existing pages), intent eligibility (never transactional primary), inventory coverage (sufficient published products), keyword ownership check
- Published with full AI content, schema, and sitemap entry

Stage 2 Post-Publish Near-Pages (5-Week Gate):

- Created exactly 5 weeks after initial batch publishing, based on real GSC emerging queries and embedding similarity analysis
- Default: noindex, follow until human approval in CMS Near-Page Control Panel
- Require explicit human approval before becoming indexable
- Must pass all Stage 1 gates plus: minimum GSC impression threshold, SERP gap confirmation, per-entity cap compliance

2.15 Continuous Optimization Loop

- Daily: GSC and GA4 signals feed the Optimization Inbox. Triggers: CTR < 1.2% with ≥200 impressions > meta refresh. Average position drops >3 over 14 days > intro regeneration. 0 impressions for 30 days > dead page queue. Bounce rate > 55% > alternative intro + highlights. Schema validation error > auto schema regeneration.
- Weekly: LangChain reclassifies all live pages into operational buckets: NEEDS_META_REFRESH, NEEDS_SNIPPET_REFRESH, NEEDS_SCHEMA_FIX, NEEDS_FACTUAL_REFRESH, HIGH_CANNIBALIZATION_RISK, READY_FOR_NEAR_PAGE_WINDOW.
- Hard rule: The optimization engine can only run allowed job types. It cannot change keyword ownership, create page types outside the near-page module, override manual locks, or auto-publish high-risk changes without queue enforcement and daily cap compliance.

System 3: Model C Decision Platform Selection Engine

Purpose

New in v2.1: Model C is the core of the Rosotravel Decision Platform. Without this system, the platform is an AI SEO engine but not a curated decision platform. This is what differentiates Rosotravel from a smaller OTA catalog.

Model C governs how products move from the raw inventory to curated page presentations. It enforces curation quality, prevents near-duplicate flooding, and produces the deterministic winner logic that powers all main-stake page blocks.

3.1 Portfolio Governance: RAW > Pool > Set

- RAW: everything ingested never shown by default. Full inventory, unfiltered.
- Pool: eligible candidates auto-added and auto-removed by eligibility thresholds (minimum quality score, content completeness, operational readiness, language coverage). Pool membership is computed automatically.
- Set: curated membership used for all main-stake page blocks requires manual confirmation by a Product Operations Lead. Stable baseline. Main stake pages use Set-only products.

Hard rule: Main stake blocks (Top Picks, Rails, Our Picks, Top Tickets, Bundles) use Set-only products with INSTANT-only eligibility gating. Browse Curated Catalog (Phase B, band-limited) may use the broader Pool.

3.2 Product Classification Engine

- Category groups: 01 tours + activities, 02 tickets + workshops, 03 transfers. Thresholds differ per group.
- Ticket-only vs tour-like classification: protects the 'Top Tickets' block from being diluted by guided tours.

- Archetype core / variant grouping: products are grouped by experience archetype. Like-with-like competition ensures winners are meaningful. Near-duplicate products within the same archetype compete only the winner surfaces in main-stake slots.
- Audience / intent / constraint / value tier tags: HARD tags (structural, from API fields duration, group size, private flag) and SOFT tags (semantic / merchandising from content analysis). SOFT tags never influence eligibility or pool membership.
- Language coverage as ranking factor: always-on language advantage rule within archetype winner selection. Products with broader language coverage rank higher within archetype when quality scores are close.
- Rosotravel Originals (This is draft name will change later) preference: Rosotravel-owned and operated products are labeled and may be pinned higher within tolerance bands of the winner selection algorithm. This preference is bounded it cannot elevate a materially weaker product above a materially stronger external product.

3.3 Winner Logic & Rails

- Best Value slot: strongest value-for-money within archetype (computed from price / quality score / inclusions ratio).
- Premium slot: highest comfort / quality execution within archetype (explicit premium signals required small group, high guide rating, comfort inclusions).
- Rails: group one or more archetype cores into a compact shortlist. Each rail surfaces one winner per archetype represented. Rails cannot surface two products from the same archetype.
- Top Tickets block: ticket-only and city pass products. Separate classification, separate block, separate winner logic.
- Top Attraction Tours: attraction-level best picks. Governed by the attraction page's Model C Set output.
- Private upgrade logic: private options are an upgrade path, not a catalog. Shown as upgrade CTA, not as a competing main-stack option. private_only constraint activates full private-only view.
- Sparse city / low inventory handling: if Set has fewer products than the minimum threshold for a block, the block is suppressed or replaced with a 'more options' CTA rather than showing near-duplicate entries.
- Bundle selection logic: bundles are first-class entities with their own Set membership. Bundle winner is selected per city by duration (1/2/3-day) from the bundle Set.

3.4 Decision Proof Content

> *New in v2.1: Every main-stake decision surface must include machine-generated but human-reviewable decision proof content. This is the core brand differentiator.*

Every key page (City, Things-to-Do, Attraction, Bundle, Near-Page) must include decision proof components. These are generated by the AI drafting pipeline Chain 2, stored in CMS as editable fields, and governed by lock rules.

- 'Why you see a shortlist' 1–2 lines explaining curation logic for this page (e.g. 'We shortlisted 4 of 23 available Colosseum tours that offer materially different experiences.')
- 'Why these picks' 2–4 bullets explaining the selection criteria applied (e.g. 'Skip-the-line guaranteed', 'Expert-led', 'Small group only', 'Covers Forum + Palatine Hill')

- 'What we skipped / trade-offs' 2–4 bullets explaining what was excluded and why (e.g. 'Audio-guide-only options no live expert', 'Large group tours with identical content')
- Reason codes: structured machine-readable tags attached to each Set product explaining why it won (e.g. top_rated, multi_language, best_value, exclusive_access, rosotravel_original). Used by Explainability panel in CMS.
- Bundle-level decision explanations: why this bundle itinerary was composed this way, what trade-offs were made.
- Attraction / POI pack decision explanations: why these 2 picks were selected from the full POI product Set.

All decision proof content is editable in CMS and must not create indexable URL variants. It is governed by the standard lock/regenerate/revert rules.

System 4: CMS & Governance Layer

Purpose

Provide Rosotravel's marketing, SEO, and operations teams with full visibility and control over every AI-generated entity, with field-level locking, keyword ownership management, batch lifecycle controls, media rights governance, and a unified Optimization Inbox.

4.1 Marketing Panel

- AI & SEO Operations Console: pipeline status, batch activation controls (Marketing Manager / Admin only), processing queue, daily cap monitoring
- Content & Structure View: per-entity content management, field-level lock/unlock/revert, semantic diff viewer (before/after), manual regeneration triggers for specific blocks
- Keyword Ownership Panel: full keyword_map view, allocation history, cannibalization risk alerts, weekly re-sync management
- Near-Page Control Panel (High Risk): Stage 1 and Stage 2 near-page candidates, similarity scores, intent eligibility, GSC signal evidence, human approval gate before indexing
- Canonical & Duplicate Inspector: pages_vectors similarity scores, canonical target management, noindex enforcement
- SEMrush Rules Panel: configurable thresholds, intent allowlists by page type, cost governance (unit caps, refresh frequency, endpoint allowlist)

4.2 Product Ops Panel

- Product Record Detail: full entity view with RAW version history, offer version, realtime offer status, classification, archetype assignment, portfolio membership flags, internal vs external rewrite lifecycle status
- Portfolio Governance: RAW > Pool > Set membership management, eligibility threshold configuration, Set flag review, winner selection audit trail, reason codes
- Algorithm Configuration: quality threshold settings, weight multipliers, language advantage thresholds all bounded, versioned, and auditable

- Derived Product Tags: HARD / SOFT tag separation clearly labeled. SOFT tags never affect eligibility.
- Explainability Panel: for every Set product, shows full reason code breakdown, archetype winner decision, and what competing products were evaluated
- Debugging Panel: full trace of pipeline steps per entity, validation failure reasons, Diff Engine decisions

4.3 SEO Governance

- Indexation control per page type with eligibility gates
- sitemap.xml, ai-sitemap.xml, and ai-summary.json managed and regenerated on every publish/unpublish
- robots.txt management, crawl budget monitoring and backpressure controls
- AggregateRating schema gate: emitted only when rating is visible in UI and schema matches displayed value exactly
- Media rights governance integrated: asset source_class, rights_status, and indexable flag exposed in CMS media library

4.4 H1-H3, Slug, Canonical & Hreflang

> *Correction from v1.0: Appinventiv will convert all H1-H3 templates, slug rules, canonical policy, and hreflang governance already defined in the original scope specification into implementation-ready engineering logic during the discovery phase. Rosotravel is not required to produce separate documentation for these they are already specified.*

4.5 Optimization Inbox & Alerts

All post-publish signals unified in one queue. Signal types: OUTDATED, MISSING, CONFLICT, LOCKED_BLOCKED_UPDATE, AI_FAILED, AI_BACKLOG, SCHEMA_INVALID, MEDIA_REVIEW_REQUIRED, EXTERNAL_API_FAILURE, NEAR_PAGE_READY_FOR_REVIEW. Every signal requires explicit human resolution. External API failures (Google Places, Wikidata, SEMrush, GSC) create EXTERNAL_API_FAILURE alerts and queue failed jobs for retry no silent SEO or indexing changes are made because of temporary API failures. Cached data is used where safe.

4.6 Roles & Permissions

Role	Key Permissions
Admin	Full system, restricted full re-run, role management, audit log, API cost governance
SEO Lead / Head of Content	Keyword ownership, canonical overrides, near-page approval (both stages), batch activation
Content Editor / Marketer	Manual edits, field locks/unlocks/reverts, optimization queue, batch QA review
Product Operations Lead	Portfolio Set management, winner rule configuration, reason code review, archetype configuration
Junior Editor	Read-only CMS, Optimization Inbox view, QA sampling tasks

System 5: Trust Layer Reviews, Human Face & Brand Expertise

Purpose

New in v2.1: The trust layer is a core platform requirement, not an optional add-on. Verified reviews power the Last Tour Snapshot, rating aggregation, destination brand expertise blocks, and the 'Human Face' proof components. Without this layer, the decision proof content has no data source.

5.1 Review Loop Foundations

- Verified booking reviews: reviews are linked to booking_id. Only booking-linked reviews are classified as verified. Verified reviews power stronger snapshot logic, schema AggregateRating, and destination brand blocks.
- Review request eligibility: after tour completion, the booking is flagged as review_request_eligible. The platform exposes this trigger to the external CRM for automated review request sequences (up to 4 messages, configurable intervals, stops on submission).
- Review source labels: every review carries source_type Rosotravel / Viator / Tripadvisor / other. Source label is displayed in UI. Imported review freshness and Rosotravel-native freshness are not treated as equivalent.
- Review object: review_id, booking_id (nullable for imported), product_id, city_id, attraction_ids[], source_type, rating, title, body, helpful_count, media_ids[], moderation_status, imported_at or submitted_at.

5.2 Review Moderation & Governance

- Moderation states: pending / approved / rejected / hidden
- Spam detection, abuse handling, duplicate detection, source integrity checks
- Customer-uploaded review media moderation (pre-publication) media flagged with linked_review_id, linked_booking_id, source_type = customer_review, moderation_status
- Negative review routing: escalation rules to be confirmed in discovery
- Review media: customer photos and videos eligible for product gallery, Last Tour Snapshot, city recent media strips only after moderation approval

5.3 Rating Aggregation

- Product-level: weighted average rating across verified reviews linked to product_id
- City-level: weighted average across verified reviews linked to city_id. Requires minimum 20 verified reviews before displaying. Source: review aggregation job, not AI estimation.
- Country-level: weighted average across verified reviews linked to products mapped to country. Requires minimum 50 verified reviews. Same source requirement.
- AggregateRating schema: emitted only when rating is visible in UI and schema value matches displayed value exactly

5.4 Last Tour Snapshot / Human Face Layer

- Recent departure window: date range of most recent tours run (rolling window). Safe wording rule: if only review/media date exists, wording references review/media freshness NOT a specific departure date unless departure data is confirmed. NOTE: (final working on this solution is to be discussed later how to show it from UX perspective to make an impression in every situation that this tour is "fresh")
- Guide / expert attribution: guide name and profile linked when Rosotravel-operated product. Not fabricated for imported products.
- Group size signal: range derived from product data, not invented.
- 'Guests loved' themes: clustered from verified review text data. Not free-form AI generation.
- Recent media strip: customer review media (Class A or confirmed reuse rights only), admin-uploaded Rosotravel media. Moderated before display.
- Fallback logic: if no recent departure data > show review-date evidence with appropriate wording. If no review data > suppress Snapshot block entirely. Never show empty or placeholder snapshot.
- Admin-managed snapshot: admins can upload and pin Snapshot media via CMS for Rosotravel-owned products.

5.5 Trip DNA (Experience Profile)

- Compact 4–5 dimension profile per product: pace / walking intensity, crowd level, learning vs sightseeing balance, comfort / logistics level, 'best for' cues
- Generated from product data, tags, and review themes not free-form AI invention
- Displayed on product pages, used in DEE-aware bundle summaries

5.6 City / Country Brand Expertise Blocks

> *New in v2.1: Destination-level brand expertise blocks are AI-generated from factual review aggregation not generic marketing copy.*

- 'About Rosotravel in {{City}}': AI-generated block describing Rosotravel's operational presence and expertise in the city. Includes city_brand_rating_value and city_brand_rating_count (from verified review aggregation, minimum 20 reviews). Includes what travelers value most (from clustered verified review themes). City-specific positioning. Versioned, lockable, regenerable.
- 'About Rosotravel in {{Country}}': same structure at country level. Minimum 50 verified reviews required before rating is displayed. country_brand_rating_value, country_brand_rating_count from review aggregation.
- Schema consistency: AggregateRating in schema matches displayed rating values exactly. Gate enforced at publishing layer.
- CMS fields per brand block: brand_expertise_enabled, brand_expertise_title, brand_expertise_body, brand_expertise_short_intro, brand_expertise_lock, brand_expertise_version, brand_expertise_generated_at

System 6: LLM Visibility Layer

Purpose

Make Rosotravel discoverable, citable, and trustworthy within ChatGPT, Perplexity, Gemini, and future LLM systems independent of Google SEO signals.

6.1 llms.txt

Markdown file at rosotravel.com/llms.txt. Entry point for LLM crawlers describes what Rosotravel is, what it does, geographic coverage, and links to all machine-readable endpoints: ai-sitemap.xml, ai-summary.json, /api/tours/feed, /openapi.yaml, MCP manifest. Updated on major structural changes. robots.txt updated to reference llms.txt.

6.2 AI Sitemap (/ai-sitemap.xml)

Entity discovery graph alongside standard sitemap. Per entry: url (canonical absolute), entity_type, primary_keyword, geo (lat/lng/geohash), isPartOf, hasPart, nearby relation fields, lastmod. Hard rule: only indexable pages are included same gate as SEO sitemap. Regenerated on every publish cycle.

6.3 AI Summary Feed (/ai-summary.json)

Compressed LLM-optimised facts per entity. Per entity: name, url, entity_type, summary_short (1–2 sentences, editorially controlled), highlights (3–7), key_faq (3–8 Q/A pairs), geo, relationships, lastmod. Deterministic no personalization, no DEE variants. Regenerated automatically on every publish cycle.

6.4 Inventory Feed (/api/tours/feed)

Full commercial catalog for bulk LLM and partner ingestion. Versioned, cursor-paginated (up to 500 per page), supports updated_since delta sync. Per tour: entity_id (stable, permanent), canonical_url, name, short_summary, highlights[], key_faq[], booking_url (Rosotravel domain only), price_from, currency (ISO 4217), availability_state (catalog-level: in_stock / out_of_stock / preorder), duration_minutes, cancellation_policy_short, curation_badges[], source_type, content_version, commercial_version.

> *Hard rule: No supplier IDs, supplier URLs, raw supplier descriptions, raw partner reviews, or restricted media fields in this feed. Violation detection excludes the item from feed and creates a blocking flag in CMS.*

System 7: MCP Server (AI Agent Query Interface)

Purpose

Read-only, tool-based interface for MCP-compatible AI agents to search inventory, retrieve tour details, and generate booking links mid-conversation.

7.1 Architecture & Safety

- MCP server reads exclusively from the published feed cache cannot query the canonical DB directly

- Cannot expose drafts, noindex entities, RAW payloads, or supplier content under any circumstance
- Protocol: JSON-RPC 2.1. Authentication: API key per registered agent. Rate limiting: configurable per key.
- All MCP requests logged: agent_id, tool_name, input parameters, response hash full audit trail

7.2 MVP Tool Set (Phase 2)

search_tours Input: country, city, categories[], max_results

Returns: entity_id, name, canonical_url, price_from, currency, duration_minutes, rating_average, source_type. Rules: published + indexable only, deterministic ordering, no DEE personalization.

get_tour Input: entity_id

Returns: full structured detail name, short_summary, highlights[], key_faq[], curation_badges[], trust signals, canonical_url, booking_url. Excludes: availability calendars, variant matrices, supplier metadata.

get_availability_link Input: entity_id, date (optional)

Returns: booking_url (signed, Rosotravel domain, always). Date is a routing hint only availability validation happens on the booking page.

get_bundles Input: city, days (1/2/3)

Returns: bundle objects name, canonical_url, highlights[] of included experiences. Same publishing gate as products.

7.3 DEE Boundary AI SOW vs Frontend Scope

> *New in v2.1: Explicit boundary table between AI SOW deliverables and Frontend/UX scope deliverables for DEE.*

AI SOW (Appinventiv) delivers	Frontend / UX Scope delivers
7 pre-written audience snippet variants per page, generated during AI drafting	Signal detection (G1–G5 signal groups, precedence logic)
CMS fields for each variant with lock, regenerate trigger, revert option	Runtime variant selection and rendering based on detected signals
Fallback rules: first_time_visitor default when no signal detected	Audience-aware block ordering on page
DEE-ready output fields in Published Record	Hook priority display logic
Intent / constraint / eligibility state tags on every product	No-new-URL behaviour enforcement
Reason codes and archetype winner data exposed via Published Record	Near-page theme seeding of DEE state on landing

System 8: Agentic Commerce API

Purpose

Allow authenticated AI agents to create priced quotes, generate signed checkout sessions, and hand off to a headless checkout enabling end-to-end AI-assisted bookings without the agent touching payment.

8.1 Quote Creation POST /api/agent/quotes

- Input: entity_id, date, pax breakdown, agent_api_key
- Output: quote_id, itemised totals, applied_rules, requirements[] schema (machine-readable list of exactly what traveller data to collect), pricing_hash, expires_at
- requirements[] schema example fields: customer.first_name, customer.email, customer.phone, traveler.dob (per traveller where required)
- Pricing hash is locked if price changes before checkout finalization, new quote required

8.2 Checkout Session POST /api/agent/checkout-sessions

- Input: quote_id + collected traveller data matching requirements[]
- Output: session_id, checkout_url (signed, time-limited, single-purpose, Rosotravel domain)
- At finalization: if sold_out > abort with error. If price_changed > reject, require new quote. UI-to-contract parity enforced.

8.3 Order Status & Cancellation

- GET /api/agent/checkout-sessions/{id}: status (created / opened / paid / expired / failed)
- GET /api/agent/orders/{id}: safe view status, totals, tour_date. No customer PII.
- GET /api/agent/orders/{id}/cancellation-link: signed self-service URL for customer. Agents cannot trigger cancellation directly in v1.

8.4 Headless Checkout UX

Checkout page reached via checkout_url: wallet-first (PayPal Express > Apple Pay > Google Pay > card fallback), no standard navigation, form fields driven entirely by requirements[] from Quote (no surprise fields), signed time-limited session. Checkout form is a hard reflection of requirements[] additional fields not in requirements[] cannot appear.

8.5 Agent / LLM Discount

Optional 5% discount (configurable) when campaign_type = agent_assistant. Quote returns price_before_discount and price_after_discount transparently. Conflict resolution: partner coupons and travel-agent trade rates take precedence. No stacking.

System 9: CRM-Ready Data Layer Foundations

Purpose

> *New in v2.1: The platform must be designed as a first-party customer data and retention architecture from the start. This is a core architecture requirement, not a later optional integration.*

The Rosotravel platform owns: event collection, booking and lead source-of-truth, consent state, identity stitching, destination and product interest data, and trigger generation. The external CRM (HubSpot or equivalent) owns: email composition, lifecycle workflows, segmentation, nurture logic, reminder sequences, campaign delivery.

9.1 Core Events Exposed

- lead_created: user submits enquiry or starts planning flow
- purchase_completed: booking confirmed with booking_id, product_id, city_id, revenue, source
- private_request_created: user submits a private tour request
- itinerary_request_submitted: user submits a custom itinerary request
- review_request_eligible: booking has passed tour date + configured delay, review not yet submitted
- abandoned_planning_flow: user reaches product page or cart but does not complete booking (configurable abandonment window)
- destination_interest_updated: user browses or saves city / attraction / product
- audience_profile_updated: DEE detects or user selects audience / intent / constraints

9.2 Identity & Consent

- anonymous_id: generated for all anonymous sessions. Persisted in first-party cookie.
- user_id: assigned on account creation or login. anonymous_id stitched to user_id on identification.
- consent_state: tracked per user and per session. Respected in all data sync and CRM automation. Consent state is never assumed explicit opt-in required.
- city_interest, product_interest, audience_profile: enrichment data exposed to CRM per user/session for segmentation and personalisation.

9.3 CRM Integration Layer

- Webhooks: lead_created, purchase_completed, private_request_created, itinerary_request_created, review_request_eligible all fired as near-real-time webhook events to CRM
- API endpoints: contact property sync, booking data sync, interest data sync
- Event sync: all core events exposed via event stream or scheduled data sync
- Hard requirement: no custom internal email automation system. Platform exposes data and triggers. CRM executes all lifecycle workflows.

System 10: External API Failover & Cost Governance

Purpose

Ensure the platform degrades gracefully when external APIs are unavailable, never makes silent SEO or indexing changes due to temporary failures, and controls API costs proactively.

10.1 Failover Rules by API

API	Failure behaviour	Governance
Google Places	Use cached factual data (TTL configurable). Queue failed enrichment jobs for retry. EXTERNAL_API_FAILURE alert in Optimization Inbox.	No schema changes on failure. No silent indexation changes.
Wikidata	Use cached entity data. Queue retry. Alert.	No schema changes on failure.
SEMrush API	Use last successful keyword retrieval. Block new retrieval until API recovers. Alert.	Unit caps enforced. Endpoint allowlist. Batching strategy. Refresh frequency controls. No repeat retrieval for unchanged entities.
GSC API	Use cached performance data. Queue retry. Alert.	Optimization loop pauses no new triggers generated from stale data.
OpenAI / LangChain	Failed drafting jobs go to retry queue (3 attempts, exponential backoff), then dead-letter queue. Publishing blocked until job resolves.	Partial failures cannot create broken indexable pages.
Copyscape	Validation held. Page queued, not blocked permanently. Alert.	Cannot skip Copyscape gate on English product pages.
Viator / OTA API	Last successful RAW snapshot retained. Ingestion queue retried. Alert.	No delta-less publishing only confirmed RAW versions proceed.

5. Out of AI Scope With Ownership Mapping

Per client feedback: every out-of-scope item now identifies where it is delivered. Nothing is left as an undefined gap.

Item	Where it is delivered	Notes
Frontend website page templates and UX rendering	Scope 3 — Global UX & Design System	Appinventiv delivers DEE-ready CMS outputs; frontend renders them
DEE runtime signal detection and rendering logic	Scope 3 — Global UX & Design System	See System 7 DEE boundary table
Payment processor integration	Chapter VI within Scope 2 — outside this AI SOW	Outside this AI SOW. Covered under a separate original scope sub-section document.
Affiliate and travel agent partner programs	Chapter VII within Scope 2 — outside this AI SOW	Outside this AI SOW. Covered under a separate original scope sub-section document.
Blog article body content writing	Client-side Rosotravel editorial team	AI scope delivers topic pool, keyword assignment, schema, meta, CMS blog panel. Article body is human-written only for E-E-A-T.
Email composition and lifecycle workflow execution	External CRM (HubSpot or equivalent)	Platform exposes events and triggers. CRM executes workflows.
CRM segmentation, nurture logic, campaign delivery	External CRM	Platform provides data layer foundations (System 9)
Mobile application development	Not in any current scope	To be scoped separately if required
Live chat and Ask Our Expert backend	Client-managed / Scope 1	AI scope delivers related tour recommendations data only
Post-migration SEO ranking analysis and backlink remediation	Client-side (Rosotravel SEO team)	Engineering delivers redirect infrastructure and migration tooling
Legacy URL mapping decisions	Client-side (Rosotravel SEO team)	Appinventiv implements redirect infrastructure; client decides URL mapping
Persistent DEE profiles and expanded near-page theme library	Phase 3 within original scope	Confirmed in scope, deferred to Phase 3

Hero video generation with Google Veo	Phase 3 (conditional) within original scope	Conditional on Viator licensing review

6. Technical Stack & Integrations

Category	Technology	Purpose
AI Content Generation	OpenAI GPT (latest)	All content drafting, FAQs, validation, translation, decision proof
AI Embeddings	OpenAI text-embedding-3-large	3072-dim vectors for all semantic operations
Pipeline Orchestration	LangChain	4-chain pipeline, clustering, weekly classification
Vector Database	Qdrant	4 collections: keywords, pages, FAQ, POI
Keyword Intelligence	SEMrush API	Volume, KD, intent, PAA questions, competitor SERP
Performance Data	Google Search Console API	Real queries, CTR, impressions, near-page triggers
Behaviour Analytics	Google Analytics 4	Bounce, dwell, conversions, optimization triggers
Factual Enrichment	Google Places API	Hours, address, coordinates, ratings
Encyclopedic Data	Wikidata API	History, sameAs, entity type, identifiers
Media Pipeline	Cloudinary	WebP, CDN, lazy-load, compression
AI Image Generation	Nano Banana Pro (Gemini)	Class A media only destinations, near-pages, missing editorial assets
Plagiarism Check	Copyscape API	English product page duplication scan
Repetition Check	Screaming Frog N-gram	Template repetition and AI footprint detection
External Product Feed	Viator / OTA API	Product and destination ingestion
Agent Protocol	MCP (JSON-RPC 2.1)	AI agent tool interface
CRM Integration	HubSpot or equivalent	External lifecycle automation, review requests, nurture

7. Deliverables by System & Phase

Deliverable	System	Phase
Viator API destination hierarchy ingestion, normalisation, automated dedup, CMS approval workflow	System 1	MVP
RAW ingestion engine — internal CMS + external OTA API	System 1	MVP
Internal vs external product rewrite lifecycle (one-time AI + lock rules)	System 1	MVP
Diff Engine with 5-domain hash-based change detection	System 1	MVP
Factual enrichment layer (Google Places + Wikidata) with factual_hash	System 1	MVP
Media source class tagging (A/B/C) + rights_status + indexable flag	System 1	MVP
Cloudinary media pipeline (Class A indexable + Class B performance-only)	System 1	MVP
Nano Banana Pro image generation for Class A editorial assets + ALT text automation + schema exposure for processed images	System 1	Phase 1.5
Keyword retrieval (SEMrush + GSC) with cost governance	System 2	MVP
Qdrant setup (4 collections) + OpenAI embedding pipeline	System 2	MVP
Keyword ownership locking + Topic Graph	System 2	MVP
LangChain 4-chain AI drafting pipeline	System 2	MVP
Schema Builder Engine (JSON-LD, stable @id, date metadata policy)	System 2	MVP
schema_map + faq_map — global ownership maps	System 2	MVP
Validation layer (GPT + Copyscape + N-gram + metadata + UI-schema parity)	System 2	MVP
FAQ pipeline (retrieval + dedup + ownership + AI answers)	System 2	MVP
Human QA sampling workflow	System 2	MVP
Atomic publishing layer (Published Record + all sitemaps + feeds + Qdrant re-embed)	System 2	MVP
Translation pipeline (NER preservation + localized URLs + lock inheritance + hreflang +	System 2	MVP

localized embeddings + CMS per-language override fields)		
Internal linking + relation outputs exposed to frontend templates	System 2	MVP
7-audience DEE variant generation (Chain 4) + CMS fields + fallback rules	System 2	Phase 2
Stage 1 immediate near-pages (batch-time generation)	System 2	Phase 2
Stage 2 post-publish near-pages (5-week gate + human approval)	System 2	Phase 2
Continuous optimization loop (daily GSC/GA4 + weekly LangChain classification)	System 2	Phase 2
Model C classification engine (RAW → Pool → Set)	System 3	MVP
Archetype core/variant grouping + winner selection logic	System 3	MVP
Intent / constraint / eligibility_state tagging on all products	System 3	MVP
Rails logic, Top Tickets, Top Attraction Tours, private upgrade logic	System 3	MVP
Sparse city / low inventory handling	System 3	MVP
Reason codes + explainability data per Set product	System 3	MVP
Decision proof content generation (why these picks, what we skipped, trade-offs)	System 3	MVP
Bundle selection logic	System 3	Phase 2
CMS Marketing Panel (all sub-panels including near-page control)	System 4	MVP
CMS Product Ops Panel (portfolio, explainability, algorithm config)	System 4	MVP
Optimization Inbox + external API failure alerts	System 4	MVP
H1-H3 / slug / canonical / hreflang rules converted to implementation logic	System 4	MVP
RBAC (5-level roles including Product Operations Lead)	System 4	MVP
Review loop foundations (verified review model + source labels + moderation)	System 5	MVP
Last Tour Snapshot / Human Face data layer (with safe wording rules)	System 5	MVP
Trip DNA (experience profile) generation	System 5	MVP

Admin-uploaded Snapshot media + stronger city/attraction aggregation + improved proof labels	System 5	Phase 1.5
Rating aggregation (product / city / country level)	System 5	Phase 2
Review request eligibility trigger (exposed to CRM)	System 5	Phase 2
City + Country brand expertise blocks (AI-generated from review aggregation)	System 5	Phase 2
Richer review theme summaries per audience + expanded near-page theme library	System 5	Phase 2
llms.txt + robots.txt update	System 6	MVP
ai-sitemap.xml + ai-summary.json + /api/tours/feed	System 6	MVP
Public API documentation (OpenAPI + MCP manifest)	System 6	Phase 2
MCP server — 3 MVP tools (search_tours + get_tour + get_availability_link)	System 7	MVP
MCP server — get_bundles tool	System 7	Phase 2
MCP server — agent-assisted planning handoff	System 7	Phase 3
Agentic Commerce API (Quote + CheckoutSession + Order endpoints)	System 8	Phase 2
Headless checkout UX (requirements[]-driven, signed URL)	System 8	Phase 2
Agent / LLM discount configuration	System 8	Phase 2
CRM-ready data layer (all core events + identity + consent + webhooks)	System 9	Phase 2
External API failover + cost governance (all APIs)	System 10	MVP

8. Assumptions & Dependencies

8.1 Client Responsibilities

- Viator API credentials and OTA API feed access provided before development begins
- SEMrush API access (Business plan or equivalent)
- Google Search Console API access for all target domains
- Google Places API key with sufficient quota for enrichment at scale
- Google Analytics 4 property access
- Cloudinary account access and sufficient quota
- Copyscape API key
- Human QA sampling (3–5% per batch) performed by Rosotravel's SEO/marketing team
- Blog article body content produced by Rosotravel's expert writers
- Product Operations Lead available from Rosotravel side for Set confirmation, winner rule review, and archetype validation
- Rosotravel to confirm or obtain written Viator permission for any derivative media usage before Phase 3 media expansion (NOTE from Client : We can not enrich the Viator media. We can only use our photos / movies but not created from their media)

8.2 Appinventiv Commitments

- Appinventiv converts all H1-H3 templates, slug rules, canonical policy, and hreflang governance from original scope specification into implementation-ready logic during discovery. No separate documentation required from Rosotravel.
- Destination hierarchy ingested from Viator API no manual file dependency
- All phasing timelines confirmed and baselined during discovery workshop before development begins

8.3 Key Risks

- LLM citation standards (llms.txt, MCP) are evolving some components may need adjustment as standards mature
- OpenAI API costs at scale for 300,000+ pages require cost modelling before batch development begins
- Near-page Stage 2 generation requires sufficient GSC signal new destinations with low traffic may not generate enough data within 5 weeks
- Viator media licensing: Phase 3 media expansion is conditional on explicit written Viator confirmation. Timeline risk if legal review is slow. (NOTE from Client : We can not enrich the Viator media. We can only use our photos / movies but not created from their media)
- Review loop data quality: Snapshot and brand expertise blocks require sufficient verified review volume before display thresholds are met
- Keyword assignment and locking rules need calibration on real data. SEMrush returns near-duplicate and sometimes wrong-language keywords, so the assignment and locking thresholds and the language filter can only be finalised after test batches on a pilot set of destinations. A calibration pass is run before scaling, and changes to these rules are governed under Section 10.3 (Change Management).

9. Acceptance Criteria

9.1 Technical Acceptance

- Destination hierarchy is ingested from Viator API and can be refreshed via API without manual file upload
- Internal products receive exactly one AI rewrite on first ingestion, then are locked for manual-only editing
- External products respect field-level lock behaviour manually edited fields are not overwritten by AI on subsequent diff events
- Diff Engine correctly classifies MINOR / MEDIUM / MAJOR on a test batch of 100 products with verified hash comparisons
- No keyword is assigned to more than one page across a test destination scope
- No two published pages in the same destination scope have cosine similarity ≥ 0.90 in pages_vectors
- No FAQ question with similarity ≥ 0.85 appears on more than one page
- All published pages pass JSON-LD validation with zero schema errors
- AggregateRating schema is absent on any page where rating is not displayed in UI
- FAQPage schema is absent on any page where FAQ block is not rendered
- Schema @id is stable across multiple re-publishes of the same entity
- Stage 1 near-pages are created during the initial batch run, not after a 5-week wait
- Stage 2 near-pages are created with default noindex,follow and require human approval before becoming indexable
- Class B (Viator-sourced) media assets are non-indexable, have no SEO alt text, and are not processed by Nano Banana
- Failed AI jobs create a dead-letter queue entry and CMS alert they never result in a partial publish
- All 7 audience snippet variants are generated and stored in CMS for each eligible page
- MCP server returns zero results containing supplier IDs, supplier URLs, or RAW payload content
- Checkout URL generated by Agentic Commerce API resolves to a valid Rosotravel domain page with price matching the Quote
- review_request_eligible webhook fires correctly after configured post-tour delay
- All core CRM events (lead_created, purchase_completed, etc.) are confirmed received by test CRM endpoint
- External API failure (simulated) results in Optimization Inbox alert, not a silent indexation change
- RAW versioning and auditability: every entity has a queryable version history given any entity_id, the system can return the full list of raw_version records, the diff classification per version, and which fields changed. Audit log is immutable.
- CMS lock / regenerate / revert behaviour: a manually locked field cannot be overwritten by any AI job. A regenerate trigger on a locked field produces a new AI draft stored as a candidate it does not overwrite the locked value until a human explicitly accepts it. A revert action restores the most recent AI-generated version of a field, clearing the manual lock.

- Human Face / Snapshot foundations: Last Tour Snapshot block is suppressed entirely when no verified review data or departure data exists no empty or placeholder snapshot is ever rendered. Safe wording is used when only review/media date exists (no reference to a specific departure date unless confirmed by departure data).
- Localized page versions are embedded into Qdrant pages_vectors in their target language after translation publish
- Product feed safety and completeness: the /api/tours/feed endpoint is validated against the following — cursor-based pagination works correctly up to 500 items per page; updated_since delta sync returns only entities changed since the specified timestamp; all required fields are present per item (entity_id, canonical_url, name, short_summary, booking_url, price_from, currency, availability_state, content_version, commercial_version); supplier-restricted field exclusion confirmed — zero supplier IDs, supplier URLs, raw supplier descriptions, raw partner reviews, or restricted media fields appear in any feed item. Violation detection correctly excludes items and logs blocking flags in CMS.

9.2 Content Quality Acceptance

- Decision proof content (why these picks, what we skipped) is present on all City, Attraction, Bundle, and Near-Page templates
- No page contains unsupported factual claims (fabricated hours, coordinates, phone numbers, prices)
- No page contains supplier raw content, supplier IDs, or supplier URLs
- Human QA sample passes on first review for at least one complete city batch

9.3 Business Acceptance (December 2026)

- ≥ 300,000 organic sessions per month (Google + LLM referrals + AI agent bookings)
- Conversion rate ≥ 1.0%
- SEO ROI ≥ 300%
- 100% of new/changed tours published within 24–48 hours (median < 24 hours)
- Zero keyword cannibalization incidents in Optimization Inbox

10. Project Governance

10.1 Roles

- Appinventiv AVP AI & Product Development: overall AI architecture ownership, client-facing delivery lead, scope governance
- Appinventiv Backend Team: pipeline development, API integrations, Qdrant, LangChain, MCP, Agentic Commerce API, CRM data layer
- Appinventiv Frontend Team: CMS panel implementation, schema injection, sitemap rendering, Published Record integration
- Rosotravel Technical Lead: Viator API ownership, destination hierarchy validation, RAW ingest reliability

- Rosotravel Product Operations Lead: Set confirmation, winner rule review, archetype validation, portfolio governance
- Rosotravel SEO / Marketing: human QA sampling, prompt feedback, brand tone governance, near-page approval

10.2 Phase Confirmation

Phase definitions, timelines, and delivery milestones are to be confirmed and baselined during the discovery workshop before development begins. This SOW defines the scope of each phase. The discovery workshop defines the schedule.

10.3 Change Management

Changes to keyword ownership rules, similarity thresholds, Model C eligibility thresholds, winner selection weights, audience taxonomy, intent/constraint taxonomy, or MCP data contract require written sign-off from both parties before implementation. Changes affecting content output at scale require a test batch run and QA sampling before production deployment.

10.4 Communication

- Weekly: pipeline progress, batch completion rates, Optimization Inbox signal volumes
- Bi-weekly: architecture decisions, API integration status, performance benchmarks
- Monthly: organic session growth, CTR trends, near-page performance, LLM referral tracking, CRM event volumes

11. Approval & Sign-Off

Client Rosotravel.com	Service Provider Appinventiv Technology
Name: ______________________________	Name: ______________________________
Title: ______________________________	Title: ______________________________
Signature: ________________________	Signature: ________________________
Date: ______________________________	Date: ______________________________